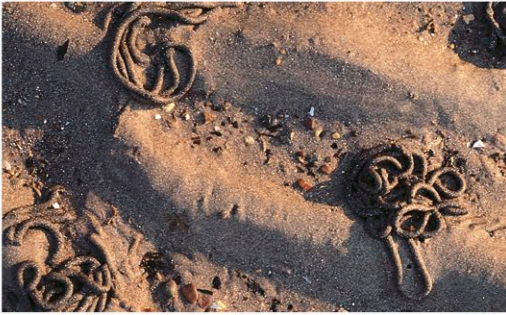




LUGWORM CASTS

Lugworm burrows have 2 holes—one for incoming water and another for ejected waste. The waste forms a pile, or “cast,” that looks like muddy toothpaste squirted out of a tube.

is an example: it makes a U-shaped burrow, lining it with mucus to keep it intact so that seawater does not make it collapse.

Mutualistic partnerships

Mutualism is a feature of life in all habitats, but it is particularly apparent on coasts. In some cases, the mutualistic partners are animals; but, in many others, one is an animal and the other is microscopic algae that live inside its body. These algae, known as zooxanthellae, can be found in thousands of coastal animals, including corals, jellyfish, and giant clams.

Zooxanthellae live by photosynthesis—the same process by which plants grow. Through a complex series of chemical reactions, they harness the energy in sunlight and use it to build up organic matter. The host animal provides the zooxanthellae with protection from the outside and, in return, the algae surrender some of their manufactured food.

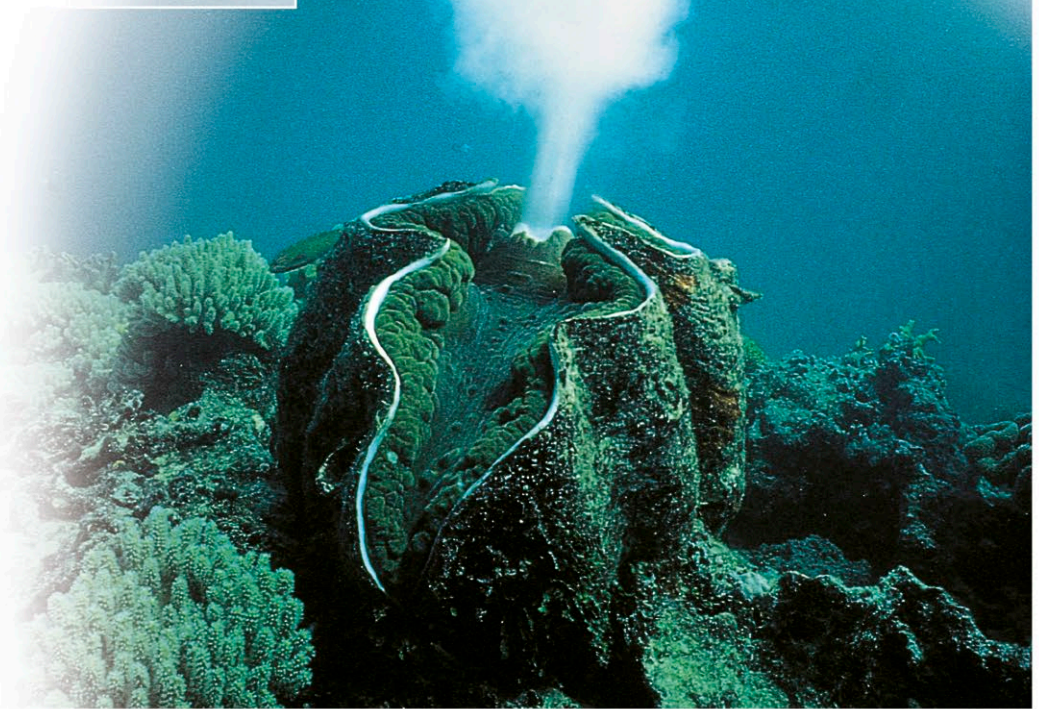
These partnerships are very important to reef-building corals because they allow them to live in places where the supply of food is otherwise low. For the partnership between

MUTUAL BENEFIT

Rather than harming clownfish, the sea anemone's stinging tentacles protect them from predators, while the fish probably help keep the anemone clean. However, the partnership is not entirely equal, for while anemones can live alone, clownfish cannot.

NEW GENERATION

A giant clam expels eggs into the sea. The larvae will drift away to settle on distant parts of the reef. Giant clams live on the food generated by the mutualistic algae harboring in their fleshy lips.

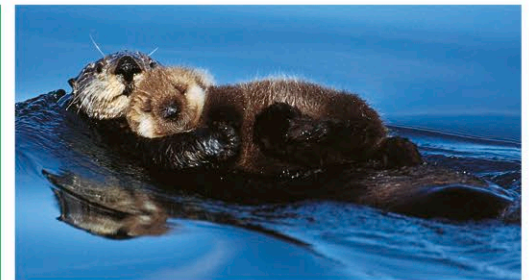


coral and algae to work, the corals have to encourage algae growth, which means they must live in bright sunshine near the water surface. However, this limits their upward growth because few corals can survive more than an hour of exposure at low tide.

Reproduction

Many land animals have adapted to life at sea. Some are now fully marine, but others—such as turtles, some sea snakes, and seals and their relatives—must come ashore to breed.

These animals are often scattered over a wide area, so they tend to form colonies during the breeding season, congregating



A LIFE IN THE SEA

The sea otter is the only otter to live entirely in water: it gives birth in the sea, and floats upside down to suckle its young. This female is swimming with her cub on her chest.

in the same place each year to maximize their chances of finding a mate. Many coastal invertebrates, on the other hand, spend their entire adult lives in one place. For them, reproduction is an opportunity not only to multiply but also to disperse. Their eggs hatch into planktonic larvae, which may then drift long distances in coastal currents before eventually settling down. For some species, such as barnacles, choosing a home is an irrevocable decision because a larva cannot detach itself once it has settled. Chemical cues help it to “make up its mind” before it takes this momentous step.

Some fish come inshore to breed because the shoreline offers plenty of hiding places for their eggs. An extreme example of such a fish is the California grunion, which lays its eggs not in water but in damp sand on beaches. Grunions stage mass spawnings during high spring tides at night. At the next spring tide, the eggs hatch and the young are washed into the sea.

